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High accuracy indoor visible light positioning using a long short term memory-fully connected network based algorithm

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Abstract: In this work, a novel positioning algorithm based on a long short term memory-fully connected network (LSTM-FCN) is proposed to improve the performance of an indoor visible light positioning (VLP) system. Using the proposed LSTM-FCN based positioning algorithm, the VLP system with a single light emitting diode (LED) and multiple photodetectors (PDs) was implemented. On this basis, the positioning performance of the established VLP system using proposed LSTM-FCN, traditional FCN and support vector regression (SVR) based algorithm is investigated and compared. It is demonstrated that the VLP system using the proposed LSTM-FCN based algorithm has better performance than that using other machine learning algorithms. As a result, an average positioning error of 0.92 cm and a maximum positioning error of less than 5 cm can be obtained for the established VLP system.

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1. Introduction

Wireless positioning systems have become an indispensable part of our daily lives over the past decade. For outdoor environment, global positioning system (GPS) has been widely used in localization, navigation and other fields. However, GPS is often unavailable in indoor environments where signals suffer from severe attenuation when passing though solid walls [1]. Nowadays, there are various indoor positioning technologies, such as radio frequency identification (RFID), Wi-Fi, ZigBee and Bluetooth. These technologies have limitations on account of low accuracy, electromagnetic interference, low security, and crowded spectrum resources [2,3]. With the development of visible light communication (VLC) based on light emitting diode (LED), visible light positioning (VLP) has attracted much attention in recent research due to its many advantages including security, license free spectrum, and safe to humans. Furthermore, the influence of multi-path reflection on positioning accuracy is not as significant as in case of radio frequency (RF) [4].

Researchers have proposed a variety of VLP techniques, including time-of-arrival (TOA) [5,6], received signal strengths (RSSs) [7,8], time-difference-of-arrival (TDOA) [9,10]. VLP technology can be further classified into non-imaging receiving positioning and imaging receiving positioning, which use the photodiode (PD) and the image sensor as the receiving terminal, respectively [11–17]. For the non-imaging receiving positioning with single PD and multi-LED, fingerprinting-based indoor localization carried out by offline phase. It has been reported that, for the fingerprinting positioning based on offline phase, a variety of RSS samples are built from different positions and stored in database as a trained set, and the location is determined by means of new RSS measurements for online phase [18]. In order to obtain higher accuracy, some new prediction technologies based on machine learning have been combined with fingerprint positioning. The k-means clustering has been combined with linear regression to achieve

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three-dimensional positioning based on the fingerprint positioning system [19]. A k-nearest neighbor-random forest (KNN-RF) model has been proposed which adopts KNN to enlarge the number of inputs [20]. A combination of machine learning classification and machine learning regression algorithms has been given to improve the positioning accuracy, in which the floor of the room is divided into a central area and an edge area [21]. Except for the traditional machine learning algorithms, neural network technology has also been applied to indoor VLP system with satisfactory results. In Ref. [22], The impact of distance metrics used to compute the weights of the weighted K-nearest neighbor (WKNN) algorithm on the localization accuracy of the VLP is investigated. And a method for estimating an optical propagation model using only a handful of measurements is developed to create a dense and accurate fingerprinting database. Zhang et al. have proposed a real-time 3D visible light positioning system using a fingerprint library and extreme machine learning [23]. Reference [24] demonstrated that receiver diversity with neural networks is very effective in providing accurate localization, and the effect of multipath reflection on positioning has been studied. Moreover, to diminish the training time and complexity, a positioning unit cell model duplication scheme, named as spatial sequence adaptation (SSA) process, was propose and demonstrate. And a residual concatenation neural network (RCNN) and transfer learning (TL) also propose and demonstrate to refine the model of the target positioning unit cell [25]. Two types of regression based machine learning algorithms, including the second-order linear regression machine learning (LRML) algorithm, and the kernel ridge regression machine learning (KRRML) algorithm were used. And a sigmoid function data preprocessing (SFDP) method was proposed to improve the positioning accuracy [26]. Reference [27] proposed a practical 3D VLP system using RSS with low complexity trilateration assisted by deep learning. And a new method of off-line preparation is adopted to minimize the workload of the VLP system deployment for more practical usage.

However, the strict requirement of LED number is more likely to lead to system failure in actual environments although the VLP system with multiple LEDs has high performance. Therefore, the positioning technology based on single LED has attracted the attention of researchers. Reference [28] has been presented a VLP system that applies received signal strength (RSS)-based fingerprinting using a weighted k-nearest neighbor (WKNN) algorithm for localization. In this work, a novel indoor positioning algorithm using long short term memory - fully connected network (LSTM-FCN) is proposed for the VLP system with single LED and multiple PDs [29]. The introduction of LSTM is expected to further improve the positioning accuracy and stability by reducing the influence of thermal noise and shot noise in the actual environment, which consistent with the characteristics of Gaussian white noise [30]. There have been some researches on the application of LSTM in noise [31–33]. Reference [31] discuss the LSTM training with noisy data following a Gaussian distribution, and demonstrate that the standard LSTM network is capable of capturing the underlying process dynamic behavior by reducing the impact of noise. The LSTM network of deep learning provides an effective way to model the nonlinear behavior of time series signals. The positioning performance of the VLP system with the proposed algorithm is further investigated and compared with that with conventional FCN and SVR [34] based algorithm.

2. Positioning algorithm and experiment setup

2.1. Positioning algorithm based on a LSTM-FCN

The positioning algorithm based on LSTM-FCN is used in this work, which structure is illustrated in Fig. 1. It can be seen that the LSTM-FCN includes a pair of LSTM formulating bidirectional LSTM (BiLSTM) and a FCN unit. As a special type of RNN, the LSTM can reads the input time series sequentially and converts the input data into a hidden state at each time step and the current hidden state is a nonlinear function of the current input and the previous hidden state [35]. The difference between LSTM and BiLSTM is that BiLSTM works in two directions: one moves forward with past data, and the other moves backward with future data. BiLSTM can

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better capture the bidirectional characteristics of data [36], the structure of an unfolded BiLSTM layer, containing a forward LSTM layer and a backward LSTM layer. The FC layer is added behind the BiLSTM layer, and the output of the BiLSTM is used as the input of the FC layer. And, the FCN is a back propagation (BP) neural network, which consists of three layers: input layer, output layer and hidden layer, and has strong generalization ability and nonlinear fitting ability. The trained LSTM-FCN can obtain the receiver's position coordinates as the output when the received power as the input.

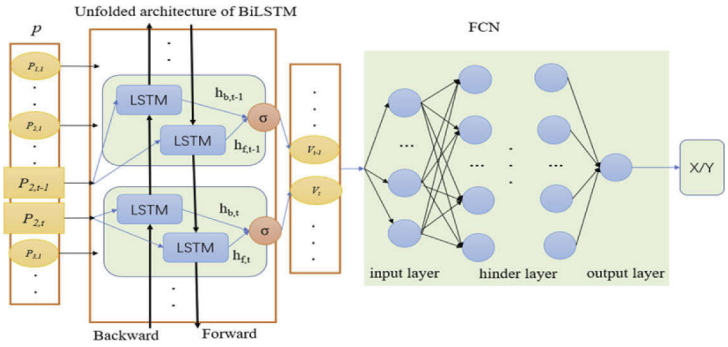


Fig. 1. The structure of LSTM-FCN used in this work for the VLP system.

The positioning algorithm based on LSTM-FCN can be divided into two phases, being online phase and offline phase, respectively. In the offline phase, $m \times n$ coordinate points used to place the receiver are selected to obtain training data. The received power at each location is recorded as:

$$p_{ij} = [\ p_{ij1} \quad p_{ij2} \quad p_{ij3}\ ] \qquad 1 \le i \le m, 1 \le j \le n, \tag{1}$$

where p_{ij1} is the received power got by the center PD, and p_{ijk} is the received power of the k-th tilted PD, which can be described as:

$$p_{ijk} = [\ p_{ijk,1} \quad p_{ijk,2} \quad \cdots \quad p_{ijk,t}\ ] \qquad k = 1, 2, 3, \tag{2}$$

where t represents that we continuously collect light intensity data at the same coordinate. The 'divide and conquer' principle, decomposing the module into multiple sub-modules, can improve system performance [24], therefore, x and y coordinates are predicted separately. One of the x-coordinate training sequence is described as:

$$TrainX_{ij} = \left[\ p_{ij} \quad x_{ij}\ \right] = \left[\ p_{ij1,1} \quad \cdots \quad p_{ij1,t} \quad p_{ij2,1} \quad \cdots \quad p_{ij2,t} \quad p_{ij3,1} \quad \cdots \quad p_{ij3,t} \quad x_{ij}\ \right], \tag{3}$$

where x_{ij} is the x coordinate of the training data. In the training process of BiLSTM, the data is input once from the beginning to the end, and then from the end to the beginning, which usually learns faster than the one-way method. For the FCN, the calculated output value using the corresponding neural network is compared with the actual output result of the training sample, and then the training error is obtained in order to minimize the cost function. After training, the LSTM-FCN for real-time positioning in the specific room environment is obtained. The criteria adopted to evaluate the results include the prediction error and the cumulative distributive function (CDF) of the error. The prediction error is defined as:

$$error = \sqrt{(x - px)^2 + (y - py)^2}, \tag{4}$$

where (x, y) is the real position and (px, py) is the estimated position.

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2.2. Experiment setup

The positioning system using the proposed LSTM-FCN based algorithm is setup experimentally. In the system, a direct current (DC) signal $x(t)$ transmitted by the single LED and received by multi-PD. The signal intensity $r(t)$ at the receiver is given as:

$$r(t) = x(t) * \left(H_{LOS} + \int H_{NLOS} \right) + n(t), \tag{5}$$

where $n(t)$ denotes the noise. At transmitter, LED light source is regarded as Lambertian radiation sources, from which the line-of-sight (LOS) channel gain can be given by [22]:

$$H_{LOS} = \begin{cases} (\frac{n+1}{2\pi})(\frac{A}{l^2})\cos^{\alpha}(\phi_1)T_s(\psi_d)G(\psi_d)\cos(\psi_d), & 0 \le \psi_d \le \text{FOV} \\ 0, & \psi_d > \text{FOV} \end{cases}, \tag{6}$$

where α is Lambertian radiant order, calculated by α = - ln(2) / ln(cos($\phi_{1/2}$)) and $\phi_{1/2}$ is the half power of the LED. A is effective area for receiving optical radiation at receiver. l is the distance between a LED and a PD. ϕ_1 is the radiation angle of LED, and ψ_d is the angle of incidence of PD. $T_s(\psi_d)$ is the gain of the optical filter, and $G(\psi_d)$ is the gain of the optical concentrator. FOV denotes the receiving field of view of the PD. The DC channel gain of the first reflection can be given by [23]:

$$H_{NLOS} = \begin{cases} (\frac{n+1}{2\pi^2})(\frac{A}{{l_1}^2{l_2}^2})\rho\cos^{\alpha}(\phi_2)dA_{ref}\cos(\lambda)\cos(\gamma)\cos(\psi_r)T_s(\psi_r)G(\psi_r), & 0 \le \psi_r \le \text{FOV} \\ 0, & \psi_r > \text{FOV} \end{cases}, \tag{7}$$

where l_1 is the distance between a LED and a reflective point, and l_2 is the distance between a receiver and a reflective point. ρ is the reflection coefficient. dA_{ref} denotes the reflective area of small region. ϕ_2 is the angle of irradiance to the reflective point, and λ is angle of incidence. γ is the angle of irradiance to the receiver and ψ_r is the angle of incidence. In this work We get T samples at an position (x, y) and record it as $R_{xy} = [r_{xy,1}\ r_{xy,2}\ldots\ r_{xy,T}]$, the noise intensity of the data is calculated using the following equation:

$$N_{xy} = R_{xy} - \frac{1}{T}\sum_{T} r_{xy,t}. \tag{8}$$

It is noted that the noise in the VLP system is shot noise and thermal noise, which amplitudes fit the Gaussian distribution power spectral density is uniformly distributed throughout the frequency range [29]. Therefore, the noise in the VLP is modeled as independent Gaussian white noise.

The block diagram of the experimental VLP system using the proposed LSTM-FCN based algorithm is shown as in Fig. 2 and the parameters used in our experiment are summarized in Table 1. Due to the powerful performance of the Raspberry Pi 4B and good support for the neural network, we choose it as the positioning terminal. However, its IO interface does not have the function of receiving analog signals, so an external ADC is needed, and the Raspberry Pi will obtain a digital signal finally. Although the Raspberry Pi can run the neural network model very well, the training of the network and the adjustment of the parameters are still a laborious task, so we choose a personal notebook with an i7 CPU for the model training, and then Ported to Raspberry Pi. The overall size of our experimental platform is 100 cm × 110 cm × 175 cm. In our experiment, 69 grid points are evenly placed in the positioning area. 42 points (blue points) are selected to form the training set, and all 69 points are positioned in the validation stage. The training data set is shown in Fig. 3. The LSTM-FC-x and LSTM-FC-y neural network models are trained for the x and y coordinates, respectively, which can has different network parameters and structures. After network training, the real-time coordinates (x, y) of receiver's position will be obtained using the LSTM-FCN and the received power as the input.

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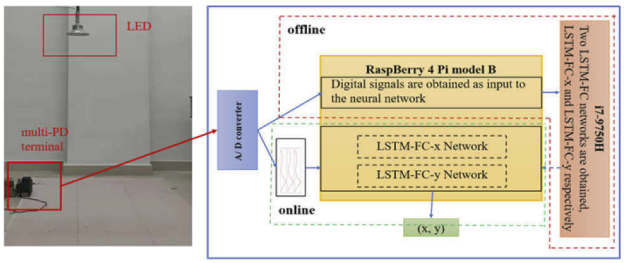


Fig. 2. The structure of the established VLP system using LSTM-FCN.

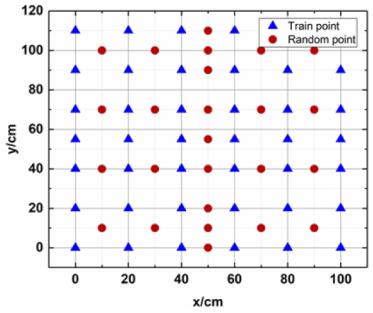


Fig. 3. Data set.

Table 1. System parameters.

Parameters	Values
Room size	1 m × 1.1 m × 1.75m
Transmitted power of the LED	20W
Field of view (FOV)	70°
Positioning terminal	Raspberry Pi 4
Number od PDs	3
Parameters of the receiver	Ref. [29]

3. Results and discussions

In this section, the performance of the VLP system using the proposed algorithm based on LSTM-FCN is further investigated. Firstly, the noise characteristics including power spectral density and amplitude distribution is given in Fig. 4. During the experiment, only one LED will be turned on, and then the multi-PD receiver will be placed in the lighting area of the LED to collect data, and then the collected data will be processed by Eq. (8). Finally, the approximate distribution of noise is calculated. It is can be seen from Fig. 4(a) that the noise power fluctuates around a certain constant in whole frequency range, which satisfies the characteristics of white noise. As shown in Fig. 4(b), the blue histogram is the distribution of the noise amplitude and the red curve is the fitted Gaussian curve. That is, it is demonstrated that the noise in VLP system is a Gaussian white noise. As a result, when the traditional positioning algorithms based on machine learning and FCN is applied, the intensity of received light signal has random fluctuations under the influence of noise However, the proposed algorithm using a LSTM layer placed in front of the FC layer can reduce the noise influence and improve the accuracy and robustness of VLP system, in which the measured light intensity of positioning point need be sampled multiple times to form a sequence. So, the LSTM layer can be used to deal with the instability of the received light

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intensity caused by the Gaussian white noise, thereby improving the performance of the VLP system.

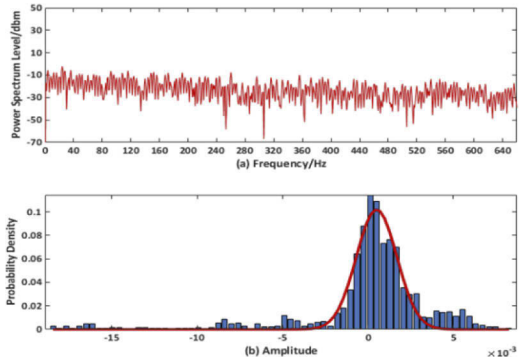


Fig. 4. Noise characteristics of the established VLP system: (a) power spectral density; (b) the distribution characteristics of the amplitude.

LSTM, a deep neural network architecture, has recently emerged as the most widely applied methodology in the analysis of sequential data. We evaluate the performance of the LSTM model by 6 groups of control experiments with light intensity sequence of different lengths, where the lengths of these sequences are 15, 30, 45, 60, 75, and 90 respectively. As mentioned above, the value of *x*- or *y*-axis of the receiver are obtained from two different LSTM-FCN models. The results are shown in Fig. 5, where the Fig. 5(a) shows the effect of sequence length on the *x*-axis, and the Fig. 5(b) denotes the y-axis, respectively. As shown in the first picture of Fig. 5(a), the horizontal axis represents the number of iterations of the parameters of the LSTM-FCN model, and the vertical axis represents the loss of the model during the training process. As the length of the input sequence increases, the *loss* parameter has a faster convergence speed. As shown in the second graph of Fig. 5(a), an increase in the length of the light intensity data sequence will bring about a better CDF curve. Figure 5(b) shows the effect of the length of the input sequence on the performance of the LSTM-FCN model on the *y*-axis. The result shows that it has the same characteristics as Fig. 5 (a). Furtrermore, Fig. 6 also shows the contribution of errors in *x*-, and *y*-axis separately to the overall performance. As the length of the input sequence increases, the y-axis positioning error will fluctuate slightly, but the overall positioning error is a downward trend. Subsequently, we summarized the positioning error of different input sequence lengths and gave the CDF curve of the error. It can be seen from Fig. 6(b) that the contribution of the error from each axis to the overall error is almost equal.

For comparison, the LSTM-FCN, FCN and SVR prediction model as a common machine learning algorithms are trained using the measured light intensity for the location plane for the established VLP system. As mentioned above, the light intensity data of location points were collected six times continuously. On this basis, using the three trained model, the location including 36 points can be predicted. Figure 7 shows the error distribution when different positioning algorithms is used, in which the black and red cylinders represent the average value and the change range of the six positioning errors, respectively. It can be seen that the VLP system using LSTM-FCN and FCN have better positioning performance than that using SVR, especially using LSTM-FCN. Moreover, when LSTM-FCN is used, the VLP system has more stable positioning results, which average errors are less than 1 cm has smaller error change as shown in Fig. 7.

In order to make a clearer comparison of positioning accuracy for different positioning algorithm, Fig. 8 gives the box diagram of error statistics and CDF results for the 69 coordinate points sampled six times. As it can be seen from Fig. 8(a), It can be seen that for LSTM-FCN

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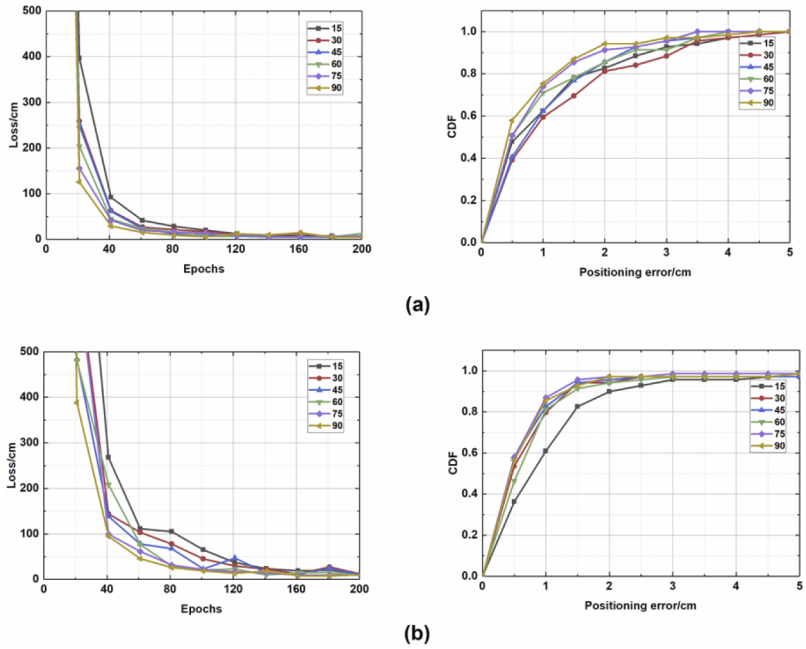


Fig. 5. Training losses and CDF of localization errors for different input sequence length. (a) *x*-axis, (b) *y*-axis.

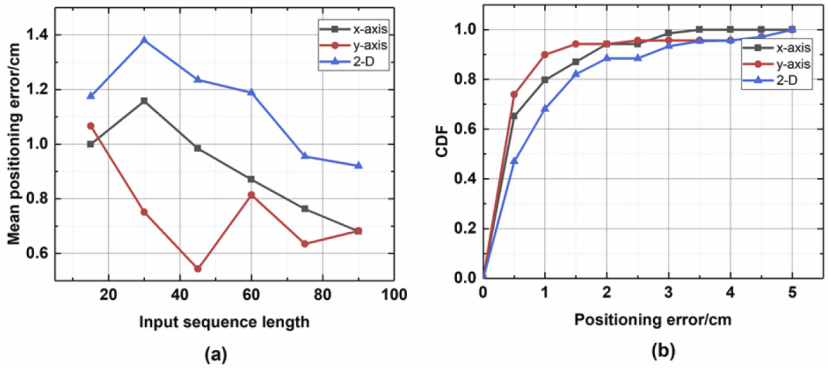


Fig. 6. Error across the different axis. (a) Mean positioning error, (b) CDF.

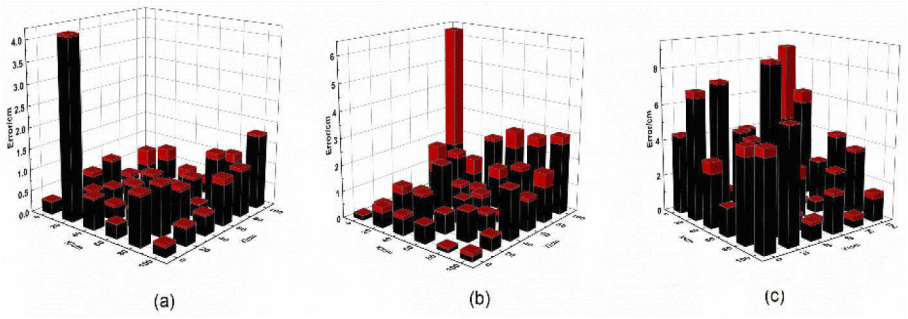


Fig. 7. The error distribution of positioning results in the location plane for the established VLP system using (a) LSTM-FCN, (b) FCN and (c) SVR model.

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based algorithm, the average error of the results is 0.92 cm, while the average error for FCN and SVR based algorithm is 2.17 cm and 3.17 cm, respectively. Moreover, there are some unsatisfactory points in the positioning results of FCN and SVR, that is, abnormal points. The reason for this is that the algorithm does not have sufficient resistance to noise. When we are offline training or online positioning, the light intensity data collected at a certain point has undesirable fluctuations. It will produce a bad result. Compared with them, LSTM no longer relies on the data of a certain time to locate, and the data sequence as the input can better resist this interference. Figure 8(b) further shows the error CDF results comparison when the three different positioning algorithms are used in the esteblished VLP system, which can more clearly present the error distribution. For the localization results using LSTM-FCN based algorithm, 70% of the points has less than 1 cm error and about 90% points has less than 2 cm error, which is consistent with the discussion obout box diagram result. For the FCN based algorithm, the system performance degrate and the error of about 40% points is less than 1 cm. But for SVR based algorithm, the error of only about 40% points is less than 2 cm. For the positioning results when SVR algorithm is used, there are even points with error greater than 20 cm. In the positioning results for FCN based algorithm, the abnormal points are improved, and the error of some points is more than 10 cm. It is demonstrated that when LSTM-FCN based algorithm in the established VLP system, the maximum error of is less than 5 cm, which reflects higher positioning accuracy and stability of positioning performance.

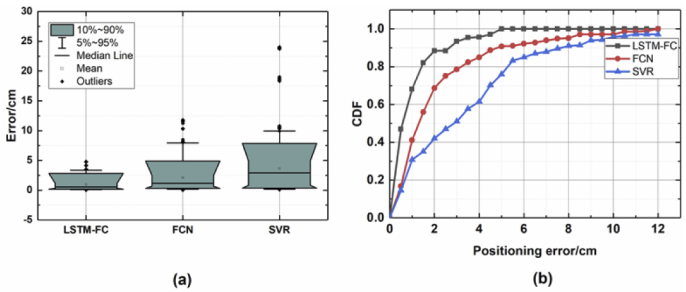


Fig. 8. Positioning accuracy comparison for the established VLP system using different algorithms. (a) BOX, (b) CDF.

Subsequently, we add a simple denoising method to the positioning algorithm based on FCN and SVR, and compare the positioning result with LSTM-FCN. One method is to average the samples before feeding the FCN or SVR. The results are shown in Fig. 9 (a), where MP is the abbreviation of mean-power. Another denoising method is to input the samples collected at the same location into the FCN or SVR one by one, and then average the estimated results. The result is shown in Fig. 9(b), where MV is the abbreviation of mean-value. Figure 9(a) and Fig. 9(b) did not show a significant performance improvement. These methods are not very good at solving the influence of noise on the algorithm model, because these data are still at the point level. For LSTM, it expands the data from the point level to the line level, which can better discover the characteristics of the data.

Moerover, the heat map of the positioning error in the 2-D plane for the established VLP system using the proposed LSTM-FCN based algorithm is given and compared with that using the traditional FCN and SVR algorithm as shown in Fig. 10. The neural network based algorithm has a larger performance improvement compared with SVR, as analyzed above. But Figs. 10(b) and (c) show that the estimation error in the edge of the positioning area is relatively high and the highest error appears in the corner. The reason for this is that as the positioning terminal moves away from the light source in the central area, the SNR decreases, and the positioning accuracy decreases due to the influence of noise. As a result, the LSTM-FCN based positioning algorithm

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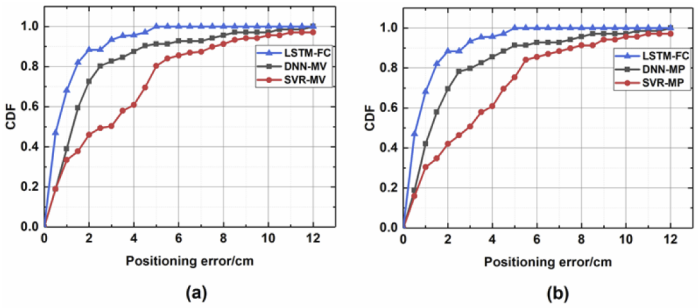


Fig. 9. FCN and SVR with simple denoising methods. (a) mean position value, (b) mean received power.

proposed in this work can well reduce the influence of noise and improve the positioning accuracy of edge areas under the same single-LED positioning system as shown in Fig. 10 (a).

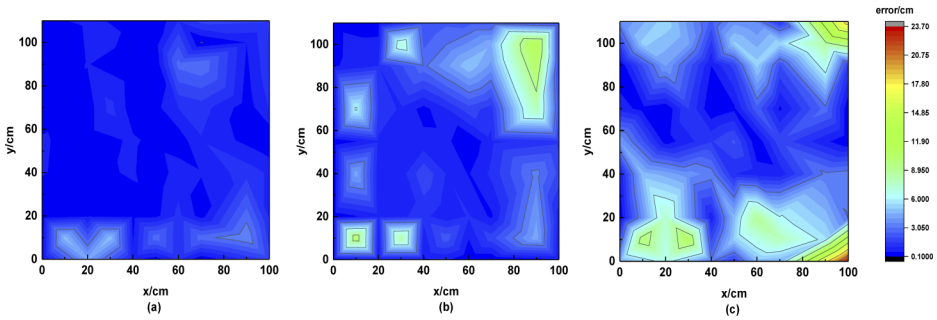


Fig. 10. The positioning results in the location plane for the established VLP system using (a) LSTM-FCN, (b) FCN and (c) SVR algorithm.

Eventually, we performed a comparative study of the proposed technique with other state-of-the-art 2D VLP techniques and summarised the results in Table 2. Though the best performance is achieved in [11], this is higher hardware complexity because of the use of the camera and the

Table 2. Comparative study of the proposed system with published work.

Ref.	Method	Accuracy (cm)	Coverage (m^2/m^3)	Notes
[39]	RSS/WKNN	1.92	2.05 × 3.05	experiment
[40]	RSS/Boosting- Bayes	9.27	10.0 × 10.0	simulation
[41]	RSS/BR-DNN	3.4	1.8 × 1.8	experiment
[22]	RSS/WKNN	2.2	3.3 × 2.1	experiment
[17]	RSS/KNN	<17	5 × 5	experiment
[12]	Camera/inertial	<16 (90%)	2.0 × 2.0	experiment
[15]	Camera	2.49	2.0 × 2.0 × 2.0	experiment
[42]	Camera	1.9	0.8 × 0.8	experiment
[11]	Camera/Bayes	0.86	1.9 × 1.0 × 1.9	experiment
[37]	magnetic and light/LSTM	<300 (87%)	2.4 × 20	experiment
[38]	inertial sensors/LSTM	78.2	128.6 (m^2)	experiment
This work	RSS/LSTM-FCN	0.92	1.0 × 1.1 × 1.75	experiment

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need for additional image processing algorithms, the program complexity is not easier than the neural network. LSTM, a deep neural network architecture, has recently emerged as the most widely applied methodology in the analysis of sequential data. LSTM networks are well suited for classification, processing, and predictions based on time-series data. As shown in [37,38], LSTM has been proposed for indoor positionin. In [37], the LSTM accurately recognizes physical activities and related action units which are generated by the inertial sensors of a smartphone, by automatically extracting the efficient features from the distinct patterns of the input data. And in [38], the smartphone receives magnetic field and light data to estimate the location of the mobile device using a probabilistic method. However, their positioning accuracy is relatively rough. The single-lamp positioning system based on LSTM-FCN proposed in this article, combined with multiple PD receivers, provides new possibilities for indoor high-precision positioning.

4. Conclusions

A novel positioning algorithm based on the LSTM-FCN is proposed for indoor VLP system with single-LED and multi-PDs. After establishing the VLP system, the noise characteristics is firstly analyzed using actual data and verified that it can be modeled as Gaussian white noise. Then the positioning performance of the VLP system is investigated and compared when LSTM-FCN, FCN, and SVR based algorithm is used. There were 69×6 positioning results, that is, 69 points were selected, and each point was positioned 6 times. The experimental results show that the LSTM-FCN has a better performance in the single-LED positioning system, with an average positioning error of 0.92 cm and a maximum positioning error of less than 5 cm. No matter the performance in global positioning or multiple positioning on a single point, it has satisfactory accuracy and stability.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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